The axial blind spot of a radial cell model: the temperature field a multi-unit model assumes, tested on the LG M50 and a 4680

Siddharth Satte, Independent researcher, Pune, India, 29 October 2026 (v1.3.0). doi.org/10.5281/zenodo.23320969. Predictions commit b92f9a8, public at 11:12 UTC on 12 October before any three-dimensional field was computed. github.com/siddharth161204-wq/m50-radial-vs-resolved-field.

1. In the multi-unit model of Yang, Cui and Tartakovsky (J. Electrochem. Soc. 173 (2026) 050520), each unit takes its temperature from a radial model, which in three dimensions means insulated ends. Under that condition the resolved cell agrees: at 2C and minus 5 C every unit lies within 0.29 K of the radial value in the LG M50 and within 1.2 K in a 4680. With the ends cooled at the paper's own h = 400 W/(m^2 K), the axial difference along the core that a radial model cannot see is 37 % of the radial core-to-surface difference in the M50 (1.28 of 3.45 K) and 32 % in the 4680 (3.44 of 10.61 K), at every rate and ambient run.
2. The radial model then puts every unit too warm: by 0.48 to 1.59 K in the M50 (mean 1.19 K, 31 of 45 units by more than 1 K) and by 1.9 to 11.3 K in the 4680 (mean 8.0 K). It overstates the unit-to-unit spread by 36 % (4.17 against 3.06 K) and by 98 % (18.9 against 9.5 K).

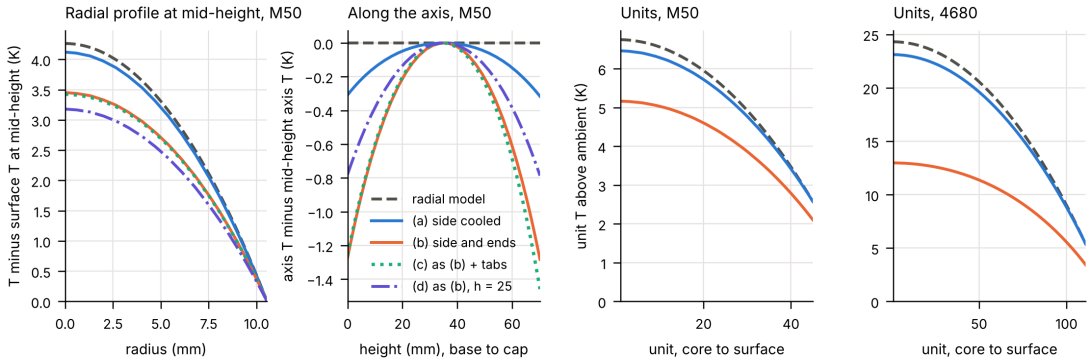


Figure 1. 2C and minus 5 C, end of discharge (1706.7 s). Left: mid-height temperature above the mid-height surface temperature, M50, the radial model against the resolved cell under the four boundary conditions. Second: axis temperature relative to mid-height, base to cap, same cases. Right, two panels: unit temperatures above ambient in the M50 and the 4680, the radial model (dashed) against the resolved annulus averages with the ends insulated (a) and cooled (b).

Method

1. Heat: PyBaMM's Doyle-Fuller-Newman model, parameter set Chen2020, discharging at 1.25C, 2C and 3C from 100 % SOC to 2.5 V at minus 5 C and 25 C, with the lumped thermal model at h = 400 on the side. The total heat is applied uniformly per volume in every thermal model, and the 4680 gets the same W/m^3.
2. Properties: O'Regan et al. (Electrochim. Acta 425 (2022) 140700) through PyBaMM at 25 C: 2938.7 kg/m^3, 842.4 J/(kg K), and 1.225 W/(m K) across and 25.09 W/(m K) along the layers. The can is 0.25 mm of steel at 15 W/(m K) and 841.7 J/(m^2 K).
3. Radial model: the paper's Eq. 2 and 4 on a solid cylinder, with 45 units (21.1 by 70.1 mm) and 111 units (46 by 80 mm) as equal-thickness annuli and the can's heat capacity at the surface. A unit's temperature is its annulus average.
4. Resolved cell: the scikit-fem cell of the two-cell study (doi.org/10.5281/zenodo.23253738), with an anisotropic roll and the can as a shell on every surface. Unit averages are taken over the annulus and the full height.
5. Boundary conditions: (a) side at h = 400, ends insulated; (b) side and ends at h = 400; (c) as (b) plus two nickel strips (6.2 mW/K each) to busbars at ambient; (d) as (b) at h = 25. The 4680 runs (a) and (b).
6. Checks:
   - With the can on the side only, the resolved cell matches the radial model within 0.15 %. This holds after the circumferential conductivity was set to the radial value, because the along-layer value failed this test by 1.7 % on the coarse mesh near the axis.
   - Energy closes to 2e-12; halving every mesh spacing changes the core-to-surface difference by 0.09 %; an independent axisymmetric solver agrees within 0.15 %.

Limits

1. The source is uniform where the paper's is unit-resolved and temperature-dependent. The heat is PyBaMM's default total, without the heat of mixing, which would add 36 to 54 % for this set. The axial-to-radial fractions are the same at every rate and ambient; the spread overstatement is 34 to 36 % in the M50 and 78 to 101 % in the 4680; the kelvin values scale with the heat.
2. The paper's properties come from its ref. 33, a pseudo-two-dimensional model with a one-dimensional thermal model of the same M50, fitted to 1C discharge data; its values were not available here. The paper's unit differences are larger than this radial model's: 2.2 and 1.7 times at 1.25C and 2C at 25 C, where both discharges end near full, and 3.5 to 3.6 times at 2C and minus 5 C, where the paper's discharge ends at 1.34 to 1.50 A h against 4.74 A h here. Its heat is larger or its conductivity lower. The roll touches the can's end caps directly, so the end effects are upper bounds.
3. No electrochemical feedback is computed here. The offsets are temperature offsets, which the paper's surrogate would turn into current and energy differences, not energy errors.

Prediction scorecard

	Prediction	Result	Computed
P1	Side cooling only: mid-height core-to-surface 3 to 10 K at 2C, minus 5 C	Held	4.12 K (central 6 K)
P2	End cooling changes it by under 30 %, axial over a third of radial	Held	-16 % and 0.37 (central 15 % and 0.5)
P3	End cooling: radial units 1 to 3 K too warm, about equally, spread within 20 %	Failed	Mean 1.19 K, all positive; the spread falls 27 %
P4	The 4680's axial-to-radial ratio is smaller than the 21700's	Held	0.32 against 0.37 (central 0.3 against 0.5)
P5	At h = 25 both differences are below 1 K at 2C	Failed	Radial 3.18 and 2.62 K, axial 0.79 and 0.65 K

What it means for the paper's numbers

Under the condition the paper's radial model implies, insulated ends, the resolved field supports it. If a cell's ends exchange heat as its side does, the unit temperatures that drive the paper's current redistribution are too warm, and their spread too wide by about a third in the M50 and by up to a factor of two in the 4680, so its 3.45 % and 11.32 % energy overestimates and its difference of more than 30 C across the 4680 would be expected to shrink, by an amount this study does not compute.

Stage 2 (v1.1.0): the gap to the paper, and an end correction

Predictions commit 0cb04c3, public at 06:37 UTC on 21 October 2026 before any Stage 2 computation; nine of ten held.

1. The paper's ref. 33 fits its thermal values to one 1C discharge of the same M50: 0.62 W/(m K) across the layers and 3.44 MJ/(m^3 K), against O'Regan's 1.23 and 2.48. They raise this model's unit differences in the M50 by a factor of 1.9 at every condition, taking the factor to the paper from 1.7 to 2.2 to 0.88 to 1.13 at 25 C, and from 3.5 to 3.6 to 1.87 to 1.91 at 2C and minus 5 C.
2. Ref. 33's electrochemical set in PyBaMM ends the 2C, minus 5 C discharge at 1.51 A h, where the paper's ends, with 1.38 to 1.40 times Chen2020's heat. With its properties and heat the remaining factor is 0.82 to 0.85 at 25 C and 1.25 to 1.33 at minus 5 C.
3. A derived, unfitted sink β (T minus T_{amb}), with $\beta = 2\ h_e/L$ and $1/h_e = 1/h + L/(6\ k_z)$, takes the case (b) unit offset from 1.19 to 0.10 K (M50) and from 7.96 to 0.21 K (4680), and the spread overstatement from 36 and 98 % to 5 and 3 %. It matches the resolved end share within 0.5 points and the axial difference at the core within 2 %.

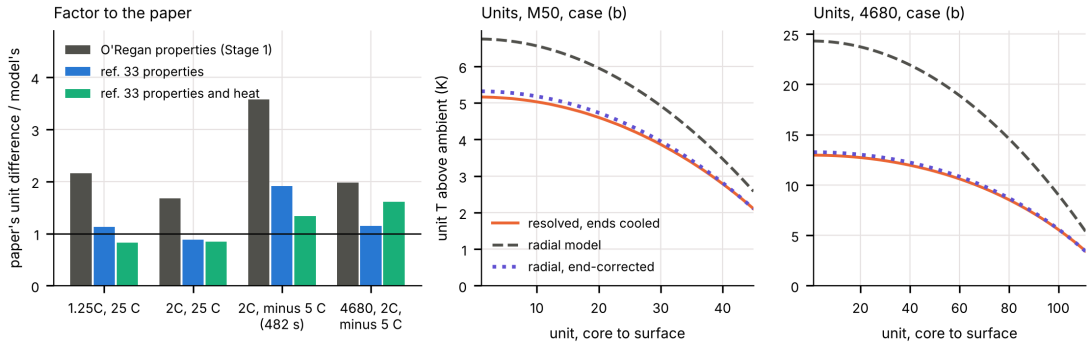


Figure 2. Left: the paper's unit difference over this model's, with each set of inputs (2C, minus 5 C at the paper's end of discharge, 482 s; the 4680 with the 21700's heat history). Right: case (b) at 2C and minus 5 C, end of discharge: resolved unit temperatures, the radial model, and the radial model with the end correction.

Stage 3 (v1.2.0): a core-temperature estimator on the radial model

Predictions commit 7060663, public at 09:36 UTC on 27 October 2026 before any Stage 3 test run; five of seven held. The two that failed: that the estimator halves the end-of-discharge miss, and that its learned error comes within 30 % of the end correction's sink.

1. Ref. 33 (Yang and Tartakovsky, J. Power Sources 664 (2026) 238948) estimates the core temperature with an ensemble Kalman filter that assimilates the surface temperature and the terminal voltage and learns the forecast's error, in the thermal equation a heat source per layer. Here the forecast is the radial model with the resolved cell's heat and properties, so its only error is the axial one, and the reading is the resolved mid-length surface temperature every 20 s, with the paper's 0.1 K noise and 3000 members; the voltage has no counterpart here. The error's noise, which ref. 33 does not give, is bracketed at 0.1, 1 and 10 times the source that moves a unit by 0.1 K in one interval, the middle level registered. Truth is Stage 1's resolved field, itself a model.
2. With the ends insulated the estimator stays within 0.14 K of the resolved core on average. With them cooled at 2C and minus 5 C it leaves 1.14 of the forecast's 1.18 K end-of-discharge miss in the M50 and 9.7 of 10.2 K in the 4680; at ten times the level, 0.48 and 5.9 K. Doubling the reading's noise adds 0.02 K to its mean error in that M50 case.
3. In the cooled M50 cases its learned error, averaged over the cell and the second half of the discharge, is 1.4 to 1.7 % of the end correction's sink on the same history, and 36 to 44 % at ten times the level. On the end-corrected forecast, at the registered level, the same estimator's mean error in those cases is 0.04 to 0.13 K, against 0.28 to 0.87 K without it, and 0.24 K against 6.6 K in the 4680.
4. For the estimator: where the radial model holds, it keeps that accuracy. Where the ends exchange heat, how much of the axial miss it removes depends on the error's noise level, which ref. 33 leaves open: at 2C and minus 5 C, almost none of the end-of-discharge miss at a tenth of the registered level, 3 % (M50) and 5 % (4680) at it, and 59 % and 42 % at ten times it. With the one-term end correction in its forecast, its end-of-discharge error at the registered level is 0.05 to 0.15 K in the cooled M50 cases and 0.31 K in the 4680.

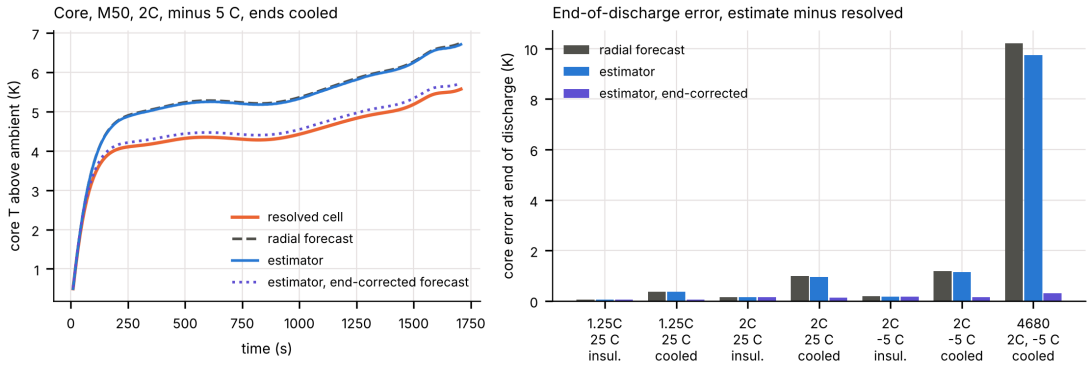


Figure 3. Left: core temperature above ambient at mid-length, M50, 2C and minus 5 C, ends cooled, registered level: the resolved cell, the radial forecast, the estimator, and the estimator on the end-corrected forecast. Right: end-of-discharge core error, estimate minus resolved, for each case and method at the registered level.

Stage 4 (v1.3.0): the current each unit draws

Predictions commit 895bd65, public at 04:06 UTC on 29 October 2026 before any Stage 4 computation; three of seven held. P3 failed (the resistance error is 1.7 %, not above 4 %), and so did P5 (the 4680's spread error is 2.6 times the M50's, not three), P6 (under Chen2020 the spread errors change by under 2 %, and the 2C, 25 C resistance error falls only to 0.54 of its value) and P7 (with the ends insulated the spread error is 7 %, not below 1 %).

1. Each unit's resistance is the cell's 10 s, 1C pulse resistance at the unit's time-averaged temperature, from PyBaMM's isothermal DFN with O'Regan's set at 100 % state of charge. At a common voltage, a unit's share of the current is 1/R over the cell's area-weighted mean; the units are equal-thickness annuli, so their electrode areas grow with radius. The share is instantaneous: the redistribution over time that the multi-unit model carries is not modelled. Minus 5 C lies below the range in which O'Regan et al. measured the kinetics (15 to 45 C) and solid diffusion (5 to 45 C), so those cases extrapolate the set.
2. With the ends cooled at 2C and minus 5 C, the radial field overstates the unit-to-unit spread of the current by 36 % in the M50 (7.6 against 5.5 % of the mean current density) and by 95 % in the 4680 (33.0 against 16.9 %), and it underestimates the cell's resistance by 1.7 and 9.6 %. The spread error tracks that of the units' time-averaged temperatures, which are 35 % too wide in the M50 and 88 % in the 4680. Under Chen2020 it changes by under 2 % (36 and 93 %), although that set's resistance is less than half as sensitive to temperature at minus 5 C. With the ends insulated it is 7 %.
3. The end correction leaves spread errors of 5 % (M50) and 3 % (4680) and resistance errors of 0.1 and 0.3 %. In the four cooled cases it removes 85 to 97 % of every error.
4. For the multi-unit model: the axial blind spot reaches the current each unit draws at the size of the thermal error. Under either parameter set, the unit-to-unit difference is about a third too large in the M50 and almost twice the true one in the 4680, while the cell's resistance moves by under 2 % in the M50. The one-term end correction brings both within about 5 %.

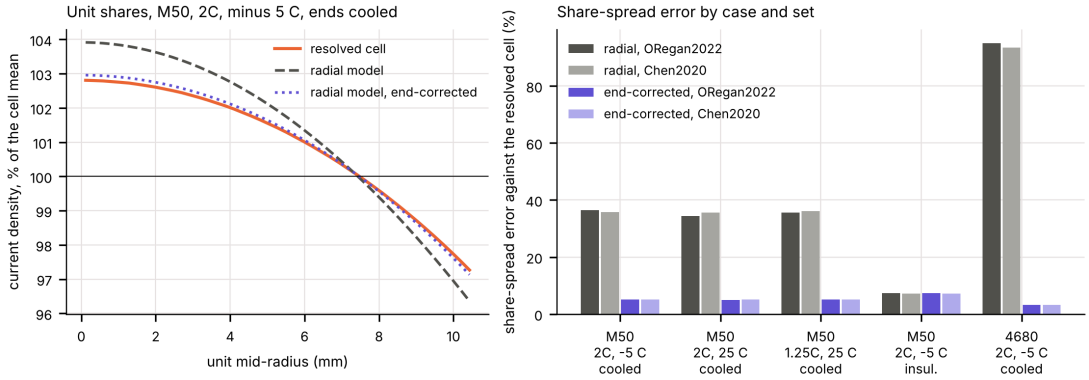


Figure 4. Left: each unit's current density as a percentage of the cell mean against its mid-radius. The case is the M50 at 2C and minus 5 C with the ends cooled, at the units' time-averaged temperatures, with ORegan2022 at 100 % state of charge, under the resolved cell, the radial model and the end-corrected radial model. Right: the share-spread error of the radial and end-corrected fields against the resolved cell, for each case under both parameter sets.